Test Management & Test Automation

Course Code: CSC4133 Course Title: Software Quality and Testing

Dept. of Computer Science
Faculty of Science and Technology

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Lecture Outline

- Test Management
- Informal & Formal Testing
- Test Team Organization
- Different Roles & Responsibilities in Major Testing Activities

- Test Automation

Objectives and Outcomes

- **Objectives**: To understand the major testing activities in the generic testing process.

- **Outcomes**: Students are expected to be able to explain the major testing activities in the generic testing process – test planning and preparation, test execution, analysis and follow-up activities.

Test Management

- People's roles/responsibilities in formal & informal testing

  - Formal testing
  - Informal testing

Test Management

- <u>In **Informal** testing:</u>
  - “run-and-observe” by testers
  - “plug-and-play” by users
  - Informal testing with *ad hoc* knowledge
  - Deceptively “easy” , but not all failures or problems easy to recognize

Test Management

- In **Formal** testing:
  - Testers, and organized in teams
  - Management/communication structure
  - Role of “code owners” (multiple roles?)
    - Dual role of developers and testers at unit/component level testing
  - 3rd party (**IV&V**) testing
  - Well-established & well-respected career path for testers
- Q: What is the career path for testers in Bangladesh?

Test Management

- Testing Teams : Organization & Management

- The test activities need to be managed by people with a good understanding of the testing techniques and processes.
- The feedback derived from analyses of measurement data needs to be used to help with various management decisions, such as product release, and to help quality improvement. Test managers are involved in these activities.
- Testers and testing teams can be organized into various different structures, but basically following either a horizontal or a vertical model. However, in reality, a mixture of the two is often used in large software organizations.
    - **Vertical** model
    - **Horizontal** model
    - **Mixed** model

Test Management

- **Basically, there are three ways to organize testers and testing teams within an IT firm:**

  1) **Vertical model** (**project oriented**) : A vertical model , would recognize around a product, where dedicated people perform one or more testing tasks for the product. *Example: one* or more teams can perform all the different types of testing for the product, from unit testing up to acceptance testing.

  2) **Horizontal model** **(task oriented):** A horizontal model is used in some large organizations so that a testing team only performs one kind of testing for many different products within the organization. *Example*: different products may share the same system testing team.

  3) **Mixed model:** Mixture of vertical and horizontal model; often used in large software organizations.

Test Management

- **External participants: Users & third-party testers**
  Besides the internal participants (just discussed in last slide), external participants or users may also be involved in testing.
  - Customers and Users may serve as testers informally for usability or beta testing
  - Independent professional testing organizations as intermediary between software vendors and customers
  - Independent verification & validation (**IV &V**) model is extensively used for certain types of software systems, e.g. defense industry or government
  - **IV&V** is getting popular. Impact of new technologies – CBSE/CBSD, COTS impact

Different Roles in Major Testing Activities

- **What are the different roles in major testing activities?**

Testing activities for large-scale testing can generally be performed and managed with the involvement of many people who have different roles and responsibilities, including

- **Dedicated professional testers** and **testing managers**
- **Developers** who are responsible for fixing problems, who may also play the dual role of testers
- **Customers** and **users**, who may also serve as testers informally for usability or beta testing
- **Independent professional testing organizations** as trusted intermediary between software vendors and customers

Test Automation

- *Flashback*: Software Testing can also be categorized as:
  - Manual Testing
  - Automated Testing
- Manual Testing: is the process of manually testing software for defects. Manual testing has few drawbacks:
  - Hard to repeat
  - Not always reliable
  - Costly
    - Time consuming
    - Labor intensive

Test Automation

- Automated Testing/ Test Automation
  - Using testing tools to execute tests with little or no human intervention
  - Testing conducted with the assistance of testing tools
- Advantages of automated testing (test automation):
  - Fast
  - Reliable
  - Repeatable
  - Programmable
  - Comprehensive
  - Reusable

Test Automation

- Basic understanding:
- Test automation aims to automate some manual tasks with the use of some software tools. The demand for test automation is strong, because purely manual testing from start to finish can be tedious and error-prone.
- Although fully automated testing is not possible, some level of automation for individual activities is possible, and can be supported by various commercial tools or tools developed within large organizations.
- The key in the use of test automation to relieve people of tedious & repetitive tasks and to improve overall testing productivity is to first examine what is possible, feasible, economical, and then set the right expectations & goals.
    - Automation needed for large systems
    - Fully automated: impossible
    - Focus on specific needs/areas

Test Automation

- **Pre-requisites for test automation**

1) The system is stable and its functionalities are well defined
2) The test cases to be automated are unambiguous
3) The test tools and infrastructure are in place
4) The test engineers have prior successful experience with automation
5) Adequate budget should have been allocated for the procurement of tools

Test Automation

- **Which Tests/Test cases to automate?**
  - Tests that should be run for every build of the application (e.g. ***regression test***)
  - Data tests that use multiple data values for the same inputs (e.g. ***data-driven test***)
  - Tests that require detailed information from the application internals (e.g., GUI attributes)
  - Tests to be used for **stress testing** or **load testing**

Bottom-line ==> Repetitive execution, better candidate for automation

Test Automation

- **Which Test/Test cases should NOT be automated?**
  - Usability testing
    - “How easy is the application to use?”
  - One-time testing
  - “ASAP” testing –”we need to test NOW!”
  - Ad hoc/random testing
    - Based on intuition, expertise, and knowledge of application
  - Tests without predictable results

Test Automation

- **<u>Question</u>: Can test automation (automated testing) *replace* manual testing?**

  - **Why or Why not ?**

Test Automation

- **Answer:**
- **No,** Test automation (automated testing) can **NOT** replace manual testing.
- Human creativity, variability, and observe-ability can not be mimicked through automation. Tools can not make decesions.
- Automation can not detect some problems that can be easily observed by a human being.
- Certain categories of tests (e.g., usability, interoperability) are often not suited for automation.
- It is too difficult to automate all the test cases.
- There will always be a need for some manual testing to some extent.

Books

- *Software Quality Engineering: Testing, Quality Assurance and Quantifiable Improvement*, by Jeff Tian

References

1. *Software Testing and Quality Assurance: Theory and Practice*, by Kshirasagar Naik, Priyadarshi Tripathy
2. *Software Quality Assurance: From Theory to Implementation*, by Daniel Galin
3. *Software Testing and Continuous Quality Improvement*, by William E. Lewis